Parallelizing Loops

Moreno Marzolla
Dip. di Informatica—Scienza e Ingegneria (DISI)
Università di Bologna

Credits

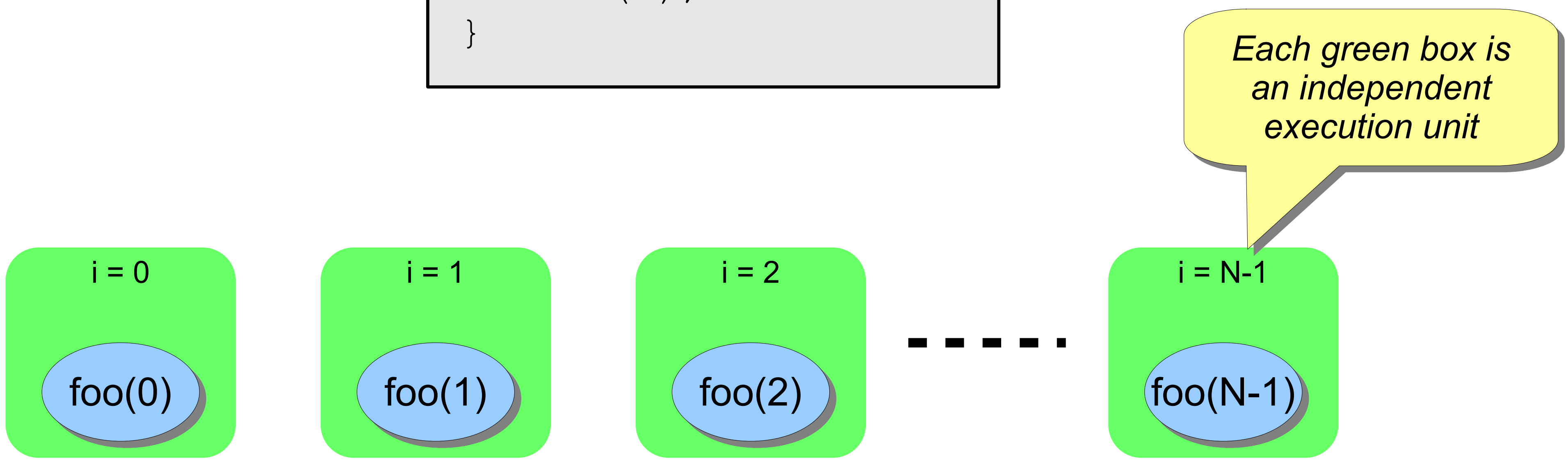
- Salvatore Orlando (Univ. Ca' Foscari di Venezia)
- Mary Hall (Univ. of Utah)

Loop optimization

- 90% of execution time in 10% of the code
  - Mostly in loops.
- Loop optimizations
  - Transform loops preserving the semantics.
- Goal:
  - Single-threaded system: mostly optimizing for memory hierarchy.
  - Multi-threaded and vector systems: loop parallelization.

Executing loops in parallel

```
for (i=0; i<N; i++) {
    foo(i);
}
```



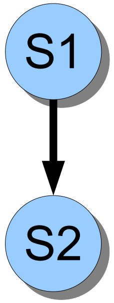
- In some cases loop iterations are independent and can be executed in parallel
  - The number of execution units could be much less than the number of loop iterations.
  - Therefore, each execution unit takes care of multiple loop iterations in some unspecified order.

Data Dependence

- Two memory accesses are involved in a data dependence if they may refer to the same memory location and one of the references is a write.

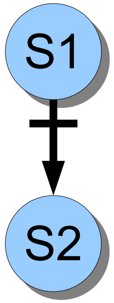
- Data-Flow or true dependence
  - RAW (Read After Write).

```
S1  a = b + c;
S2  d = 2 * a;
```



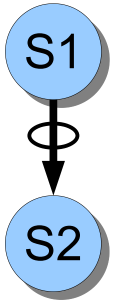
- Anti dependence
  - WAR (Write After Read).

```
S1  c = a + b;
S2  a = 2 * a;
```



- Output dependence
  - WAW (Write After Write).

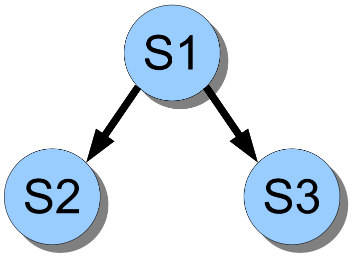
```
S1  a = k;
    if (a>0) {
S2      a = 2 * c;
    }
```



Control Dependence

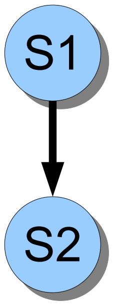
- An instruction S2 has a control dependence on S1 if the outcome of S1 determines whether S2 is executed or not.
  - Of course, S1 and S2 can not be exchanged.
- This type of dependence applies to the condition of an if-then-else or loop with respect to their bodies.

```
S1  if (a > 0) {
S2          a = 2 * c;
    } else {
S3          b = 3;
    }
```



Dependence

- In the following, we use a simple arrow to denote any type of dependence.
  - S2 depends on S1.

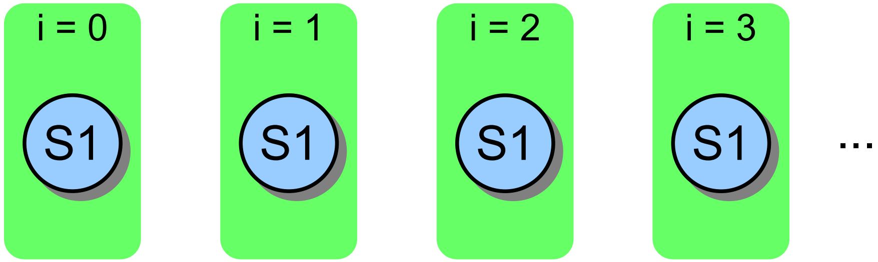


Fundamental Theorem of Dependence

- *“Any reordering transformation that preserves every dependence in a program preserves the meaning of that program”*
- Recognizing parallel loops (intuitively):
  - Find data dependences in a loop.
  - No dependences crossing iteration boundary → loop can be parallelized.

Example

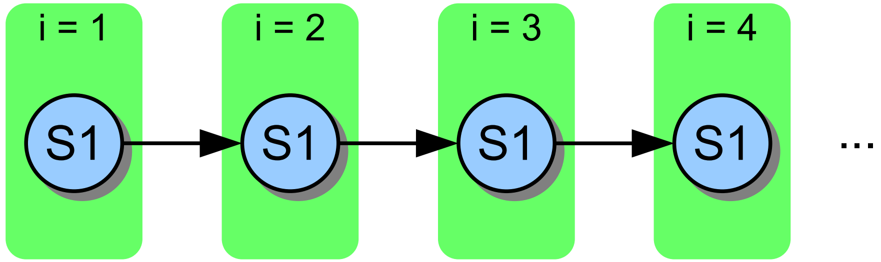
```
for (i=0; i<n; i++) {
S1 a[i] = b[i] + c[i];
}
```



- Each iteration does not depend on previous ones.
  - There are no dependences crossing iteration boundaries.
- This loop is fully parallelizable.
  - Loop iterations can be performed in parallel, in any order.
  - Iterations can be split across processors.

Example

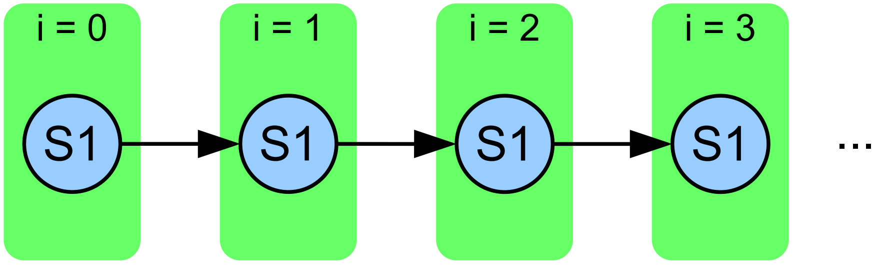
```
for (i=1; i<n; i++) {
S1 a[i] = a[i-1] + b[i]
}
```



- Each iteration depends on the previous one (RAW).
  - The value a[i-1] read at iteration *i* has been modified at iteration i-1;
  - Loop-carried dependence.
- Hence, this loop is not parallelizable with trivial transformations.

Example

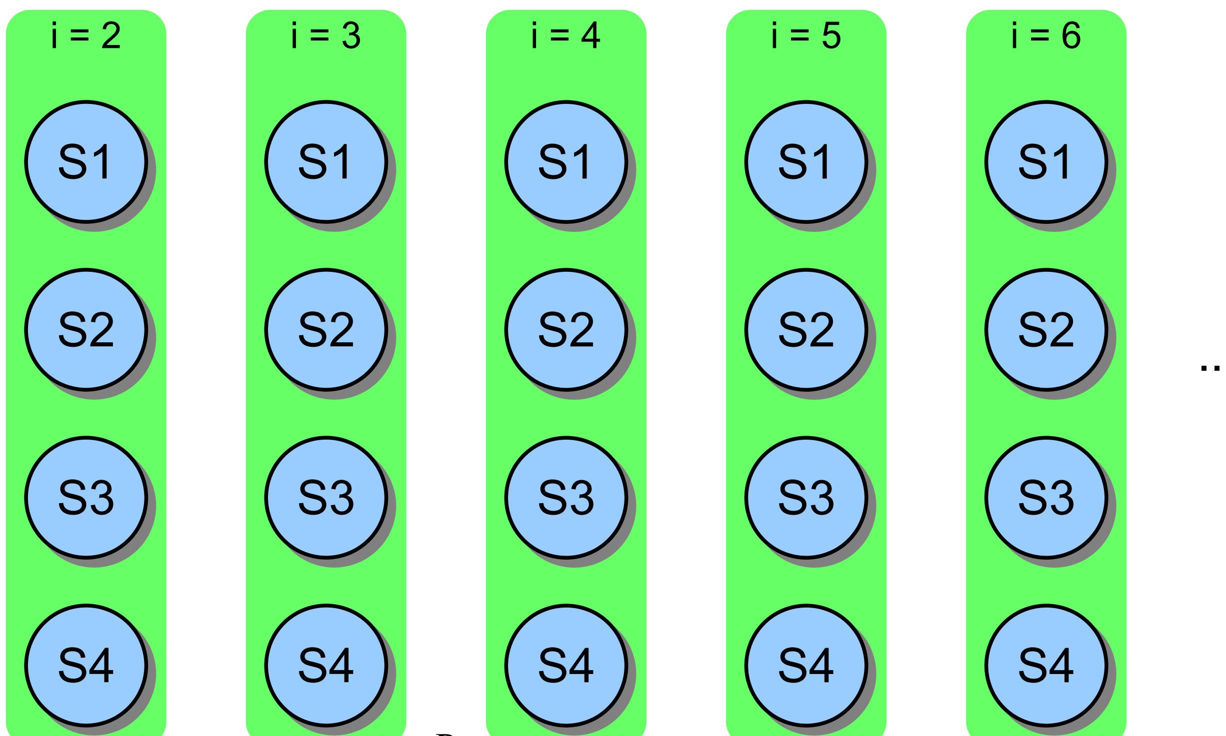
```
s = 0;
for (i=0; i<n; i++) {
S1 s = s + a[i];
}
```



- We have a loop-carried dependence on s that can not be removed with trivial loop transformations...
- ...but can be removed with *non-trivial* transformations.
  - This is a reduction, so we can apply the reduction parallel pattern.

Exercise: draw the dependences among iterations of the following loop

```
for (i=2; i<n; i++) {
S1  a[i] = 4 * c[i-1] - 2;
S2  b[i] = a[i] * 2;
S3  c[i] = a[i-1] + 3;
S4  d[i] = b[i] + c[i-2];
}
```



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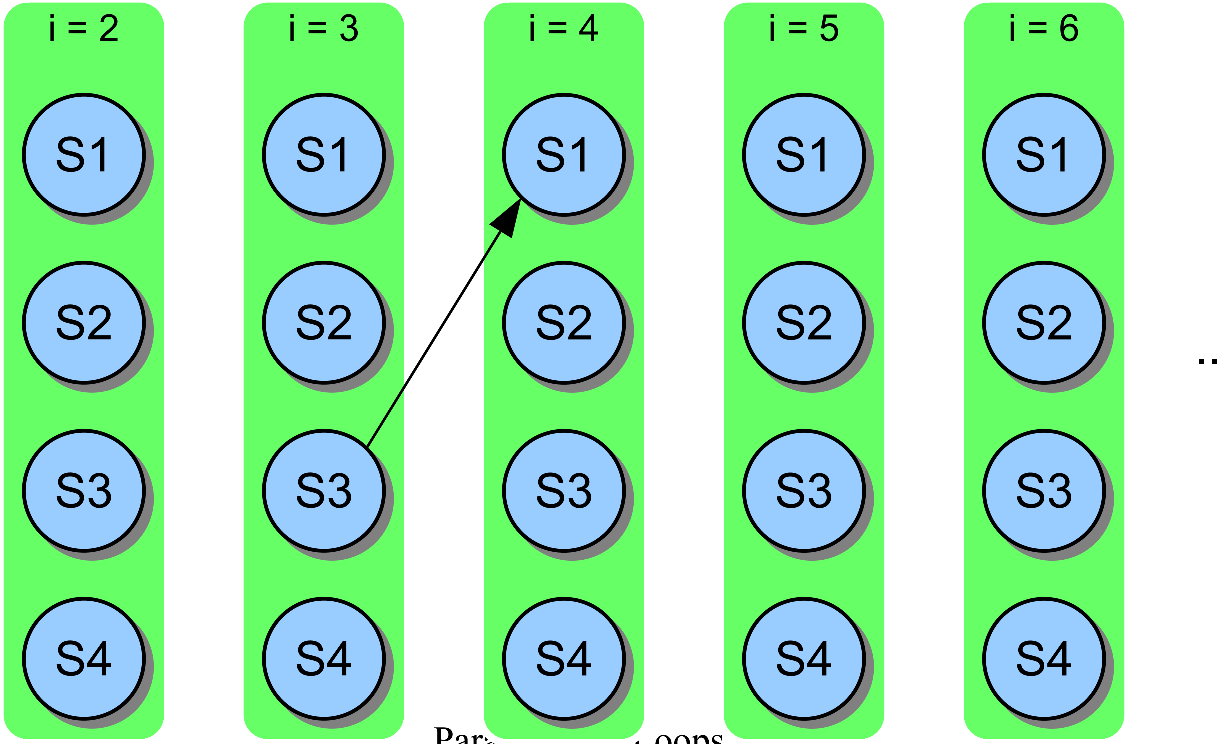

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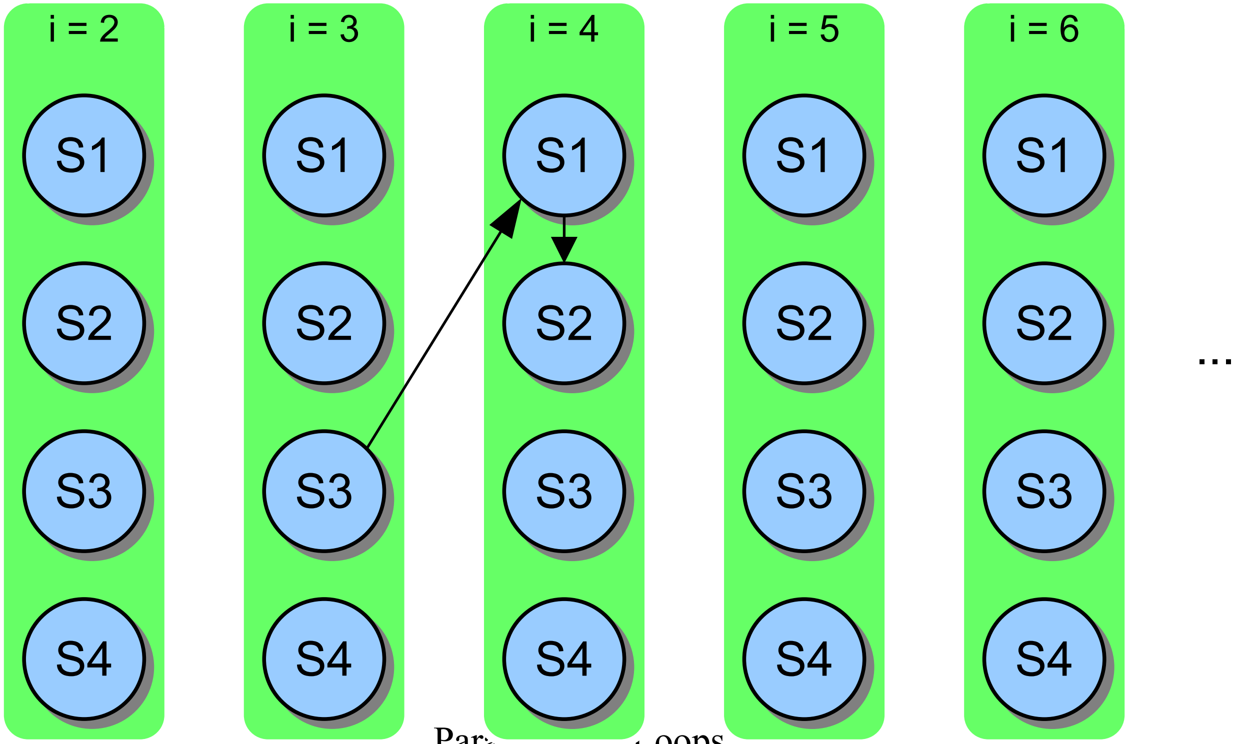

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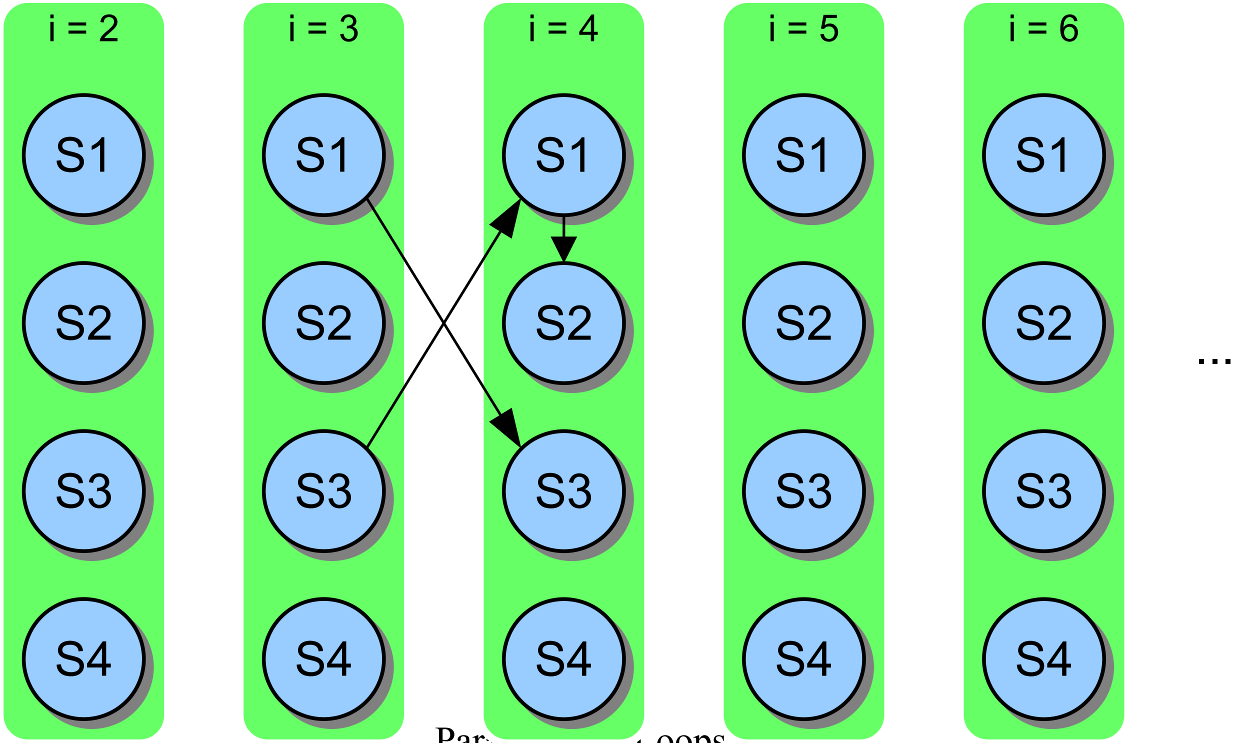

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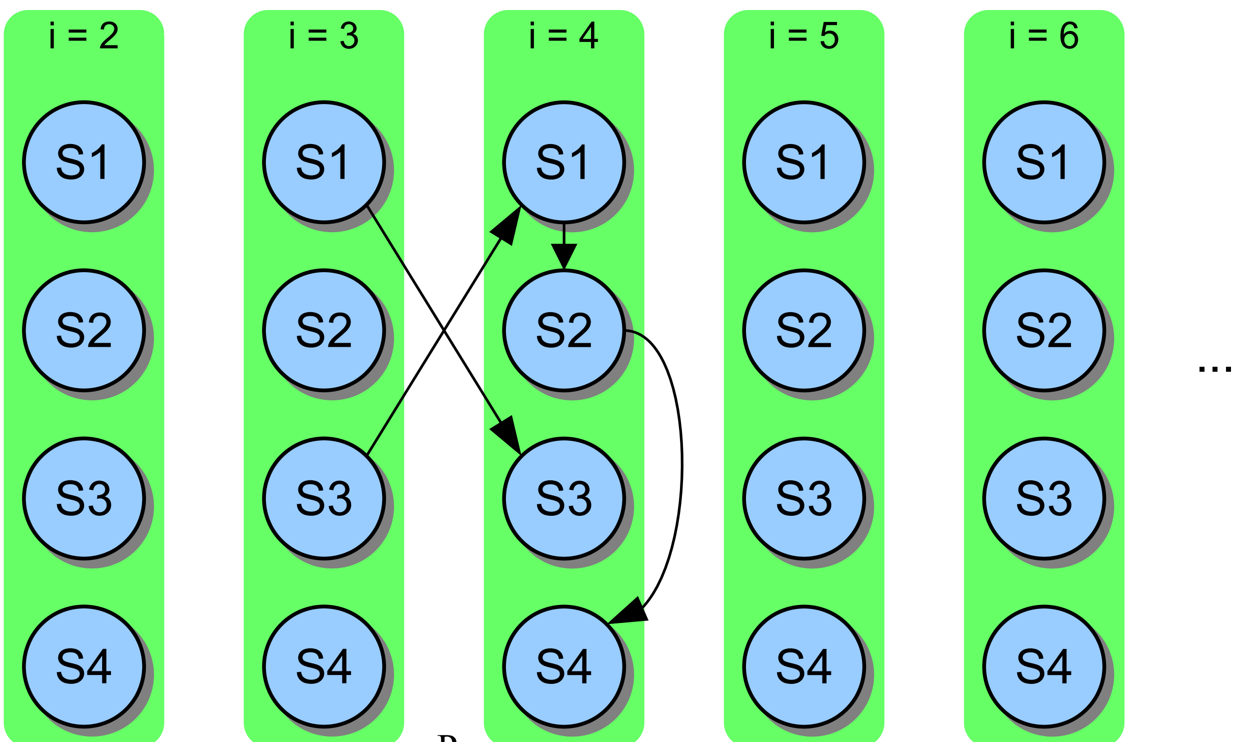
S4 d[i] = b[i] + c[i-2];
}
```



Exercise: draw the dependences among iterations of the following loop

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for (i=2; i<n; i++) {
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}
```

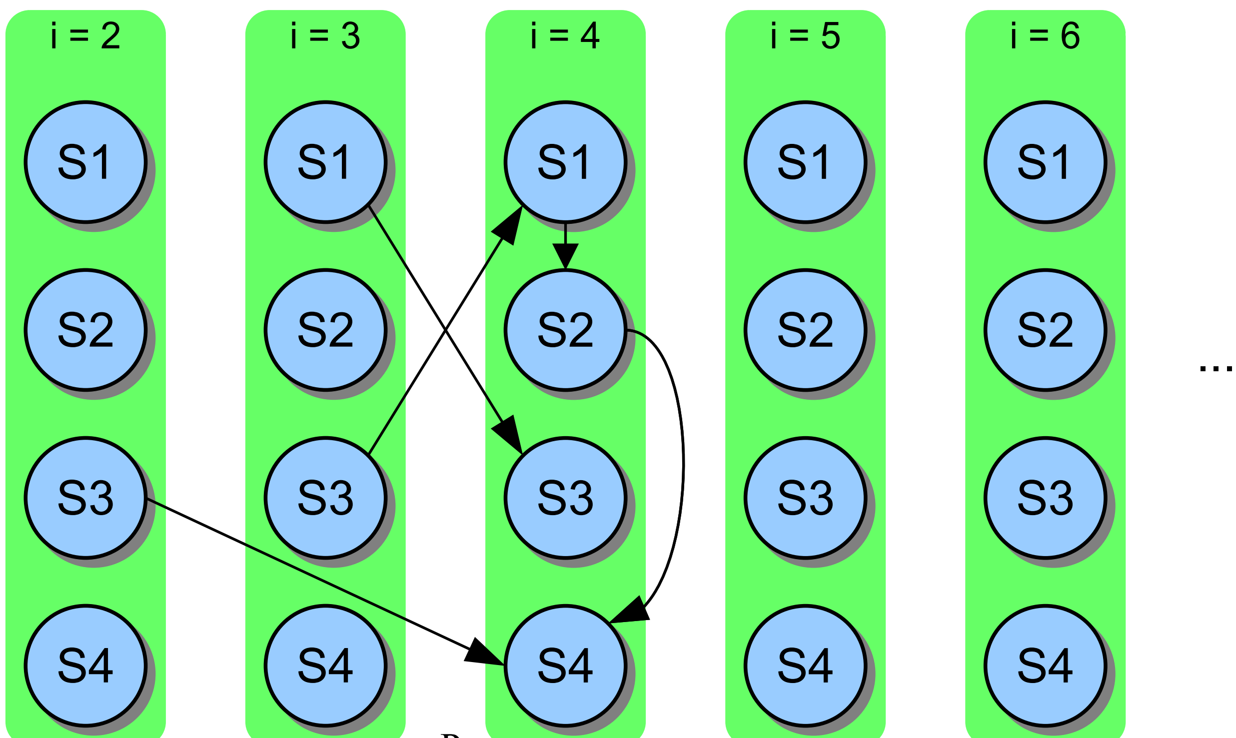

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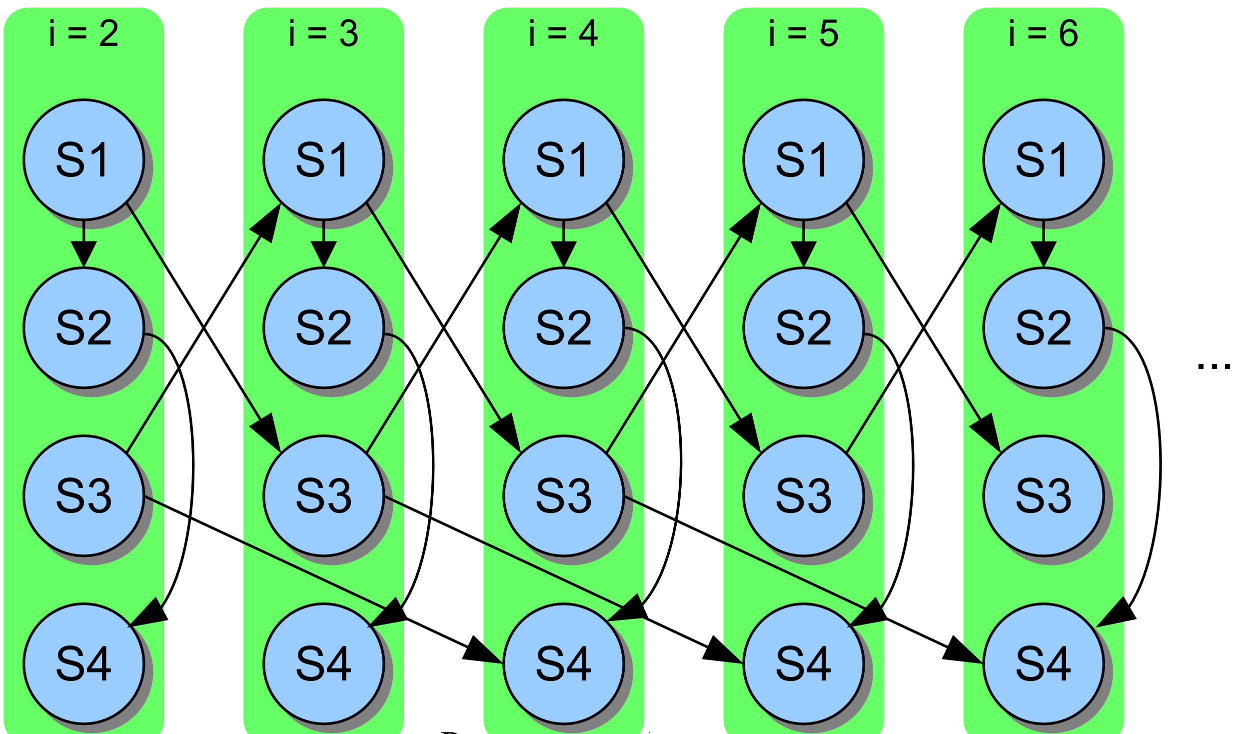
}
```

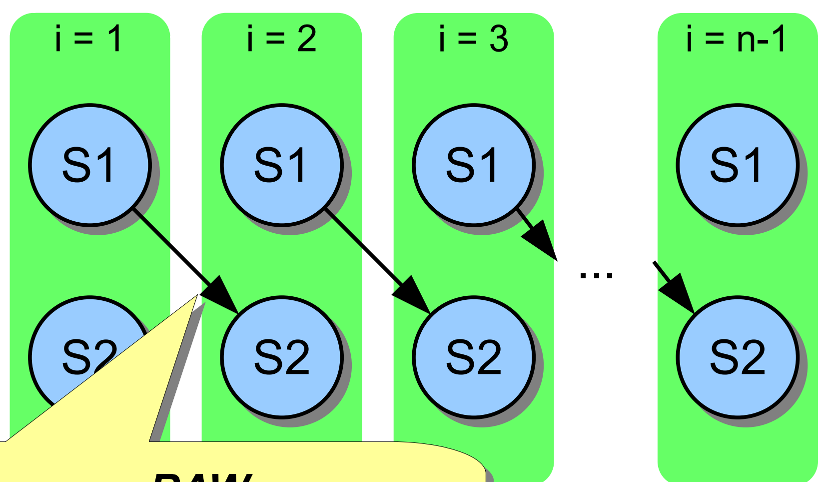


Removing dependences: Aligning

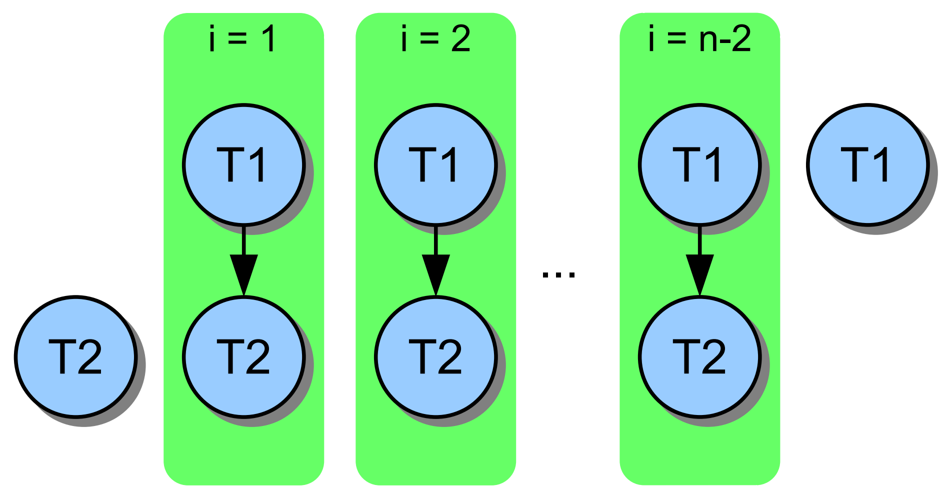
- Dependences can sometimes be removed by aligning loop iterations.

```
a[0] = 0;
for (i=1; i<n; i++) {
S1  a[i] = b[i-1] * c[i];
S2  d[i] = a[i-1] + 2;
}
```

```
a[0] = 0;
d[1] = a[0] + 2;
for (i=1; i<n-1; i++) {
T1  a[i] = b[i-1] * c[i];
T2  d[i+1] = a[i] + 2;
}
a[n-1] = b[n-2] * c[n-1];
```



RAW
a[i-1] is read at iteration i and written at iteration i-1.

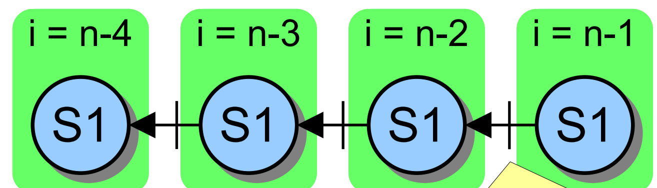


Removing dependences: Fission

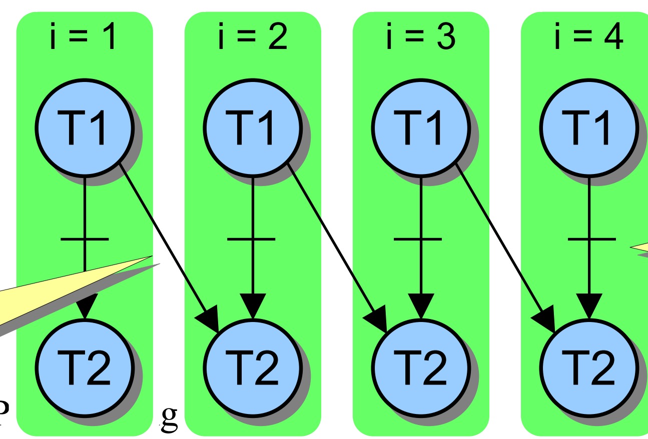
- In this example, dependences can be removed by using a temporary array and splitting the loop.

```
for (i=n-1; i>0; i--) {
S1  a[i] = a[i-1];
}
```

```
b[0] = a[0];
for (i=1; i<n; i++) {
T1  b[i] = a[i];
T2  a[i] = b[i-1];
}
```



Write-After-Read (WAR)
a[i] is written at iteration i and read at iteration i+1.



RAW
b[i-1] is read at iteration i and written at iteration i-1.

WAR
a[i] is written at iteration i after it is read.

Removing dependences: Fission

- In this example, dependences can be removed by using a temporary array and splitting the loop.

```
for (i=n-1; i>0; i--) {
S1  a[i] = a[i-1];
}
```

```
b[0] = a[0];
for (i=1; i<n; i++) {
T1  b[i] = a[i];
T2  a[i] = b[i-1];
}
```

Both loops can be parallelized

```
for (i=0; i<n; i++) {
    b[i] = a[i];
}
for (i=1; i<n; i++) {
    a[i] = b[i-1];
}
```

Optimizing loops: Fusion

- Sometimes it is possible to merge multiple parallelizable loops to reduce overhead.

```
for (i=0; i<n; i++) {
    a[i] = b[i] * c[i];
}
for (i=0; i<n; i++) {
    b[i] = f(d[i]) * h[i];
}
```

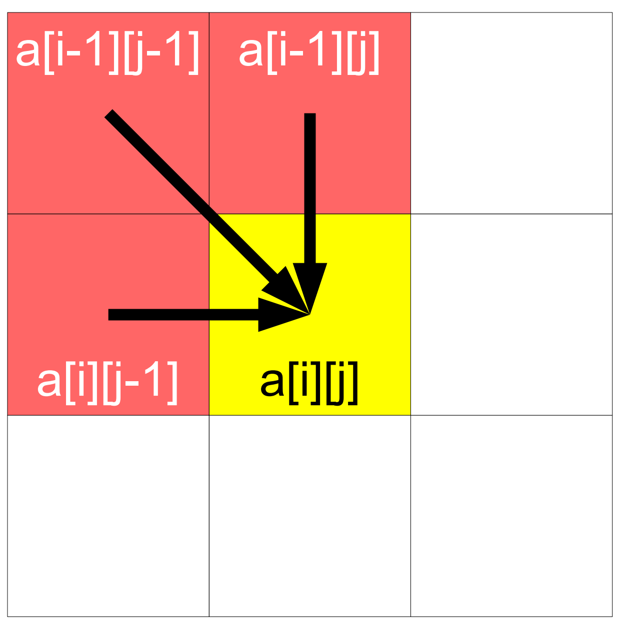
Both loops are parallelizable, since they have no loop-carried dependences.

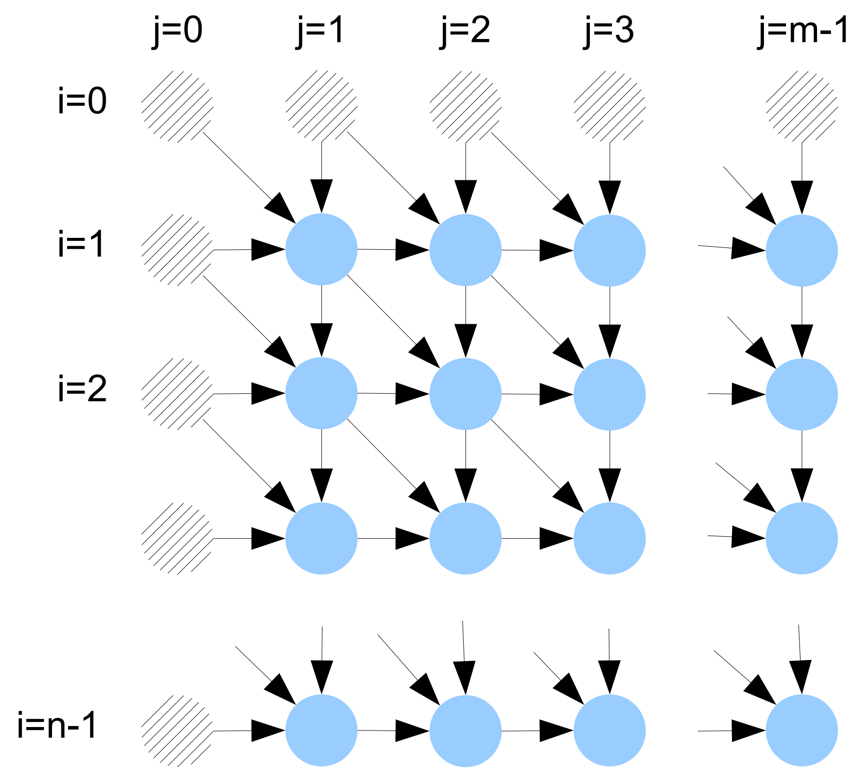
```
for (i=0; i<n; i++) {
    a[i] = b[i] * c[i];
    b[i] = f(d[i]) * h[i];
}
```

The merged loop is equivalent to the one above, and has no loop-carried dependences either. So it can be parallelized.

Difficult dependences

```
for (i=1; i<n; i++) {
  for (j=1; j<m; j++) {
    a[i][j] = a[i-1][j-1] +
              a[i-1][j  ] +
              a[i  ][j-1];
  }
}
```

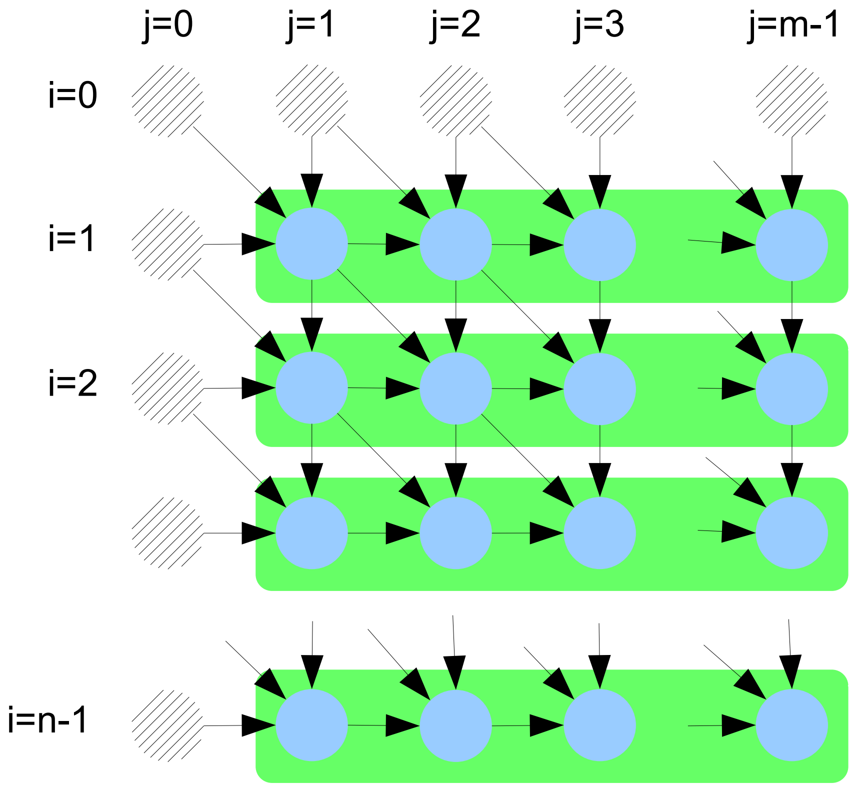




Difficult dependences

- It is not possible to parallelize the loop iterations no matter what loop we consider

```
for (i=1; i<n; i++) {
  for (j=1; j<m; j++) {
    a[i][j] = a[i-1][j-1] +
              a[i-1][j  ] +
              a[i  ][j-1];
  }
}
```



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- It is not possible to parallelize the loop iterations no matter what loop we consider

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              a[i-1][j  ] +
              a[i  ][j-1];
  }
}
```
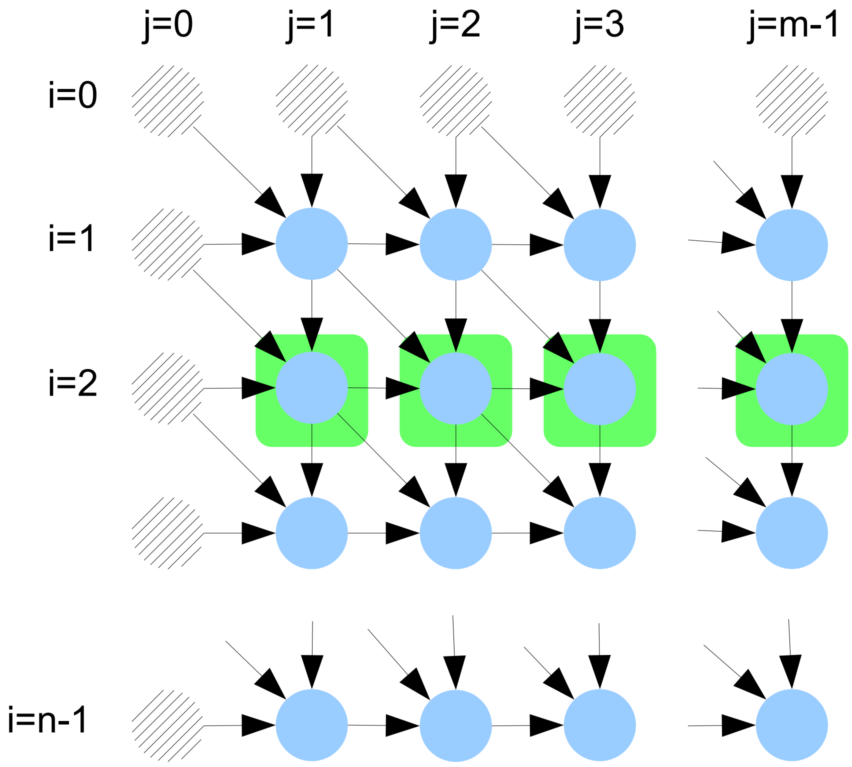
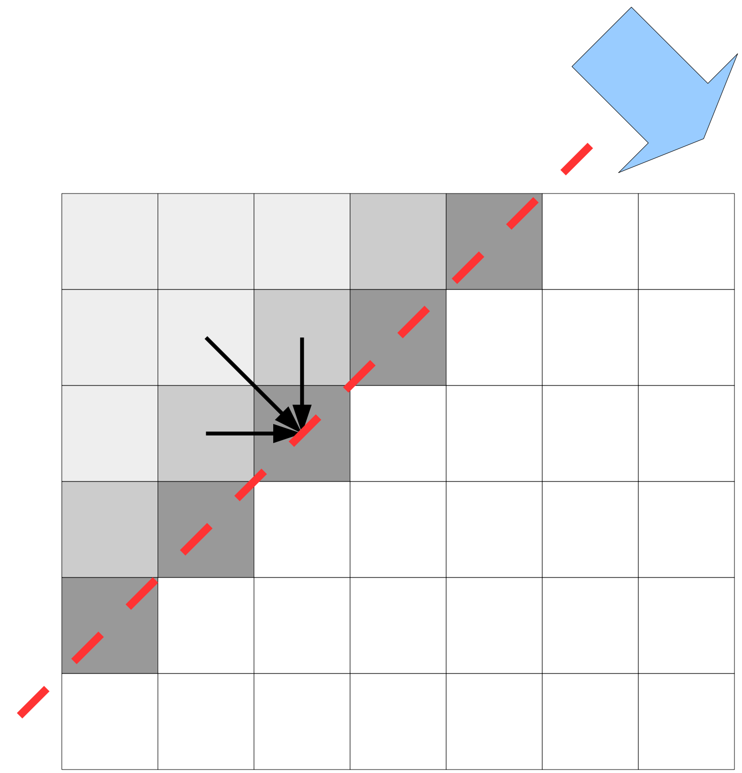



Solution

- It is possible to parallelize the inner loop by sweeping the matrix diagonally
  - *Wavefront sweep*



```
for (slice=0; slice < n + m - 1; slice++) {
    z1 = slice < m ? 0 : slice - m + 1;
    z2 = slice < n ? 0 : slice - n + 1;
    /* The following loop can be parallelized */
    for (i = slice - z2; i >= z1; i--) {
        j = slice - i;
        /* process a[i][j] … */
    }
}
```


https://stackoverflow.com/questions/2112832/traverse-rectangular-matrix-in-diagonal-strips

Exercises

Exercise

- Can this loop be parallelized? How?

```
#define N some_big_number
double a[N];
const double f = ...;
double val = 1.0;
int i;

for (i=0; i<N; i++) {
    a[i] = val;
    val = val * f
}
```

Exercise

- Which loop(s) (if any) can be parallelized? Why?

```
#define N some_big_number
int s[N] = {1, 1, ... 1};
double p[N] = { ... };

/* Assume that function f() has no side effects */

for (int i=0; i<N; i++) {
    if (s[i]) {
        for (int j=0; j<N; j++) {
            if (s[j] && f(p[i], p[j])) {
                s[j] = 0;
            }
        }
    }
}
```

Exercise

- Which loop(s) can be parallelized? Why?

```
#define N 10000
double phi[2][N][N], maxdelta;
const double EPS = 1.0e-6;
int cur = 0, next = 1;

/* ...Initializations not shown... */

do {
  maxdelta = 0.0;
  for (int i=1; i<N-1; i++) {
    for (int j=1; j<N-1; j++) {
      phi[next][i][j] = (phi[cur][i+1][j] + phi[cur][i-1][j] +
                         phi[cur][i][j+1] + phi[cur][i][j-1]) / 4;
      const double delta = fabs(phi[next][i][j] - phi[cur][i][j]);
      if (delta > maxdelta) {
        maxdelta = delta;
      }
    }
  }
  /* exchange "cur" and "next" */
  const int tmp = cur; cur = next; next = tmp;
} while (maxdelta > EPS);
```

Conclusions

- A loop can be parallelized if there are no dependences crossing loop iterations
- Some kinds of dependences can be eliminated by
  - applying code transformations (*reordering*, *aligning*)
  - using patterns (e.g., *scan*, *reduction*)
  - sweeping across iterations "diagonally"
- There are cases where dependences can simply not be removed